भारतीय सूचना प्रौद्योगिकी संस्थान, पुणे
Indian Institute of Information Technology, Pune
(An Autonomous Institute of National Importance under act of Parliament)

Mid-Term Examination
(BT13106) Software Engineering

Course: T. Y. B. Tech (Semester V)
Duration: 1:30-hour
Date: 29/10/2025

Branch: CSE
Max. Marks: 30

Student MIS no.: 112315156

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate maximum marks.
3. Write your MIS Number on Question Paper.
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1. What are the key outputs during an iteration in a project following an iterative development model? **(3 marks)**
2. What are the main components of an SRS? What are the main criteria for evaluating the quality of an SRS? What are the main error types for requirements? **(2+2+2 marks)**
3. How does prototyping contribute to requirement validation and reduction of project risks? Illustrate with an example. **(3 marks)**
4. Choose any online social networking site (eg. LinkedIn). Define its key use cases along with relevant actors, conditions, and exception flows, and illustrate them through a comprehensive Use Case Diagram. **(3+3 marks)**
5. A healthcare company is developing an AI-powered Smart Health Monitoring System (SHMS) for home patients. Patients wear IoT-enabled devices that continuously monitor vital signs such as heart rate, blood pressure, and oxygen levels. The system analyzes real-time data using AI algorithms to detect anomalies and predict potential health risks. Alerts are automatically sent to patients and doctors when readings cross critical thresholds. Doctors can review patient history, recommend interventions, and update treatment plans remotely. Based on this scenario, develop a Sequence Diagram showing interactions between patients, IoT devices, AI monitoring engine, and doctors, including exception handling for abnormal readings. **(4 marks)**
6. What challenges do large organizations face in adopting Agile methods, and how do factors like structure, culture, processes, and project scale make the transition difficult? **(3 marks)**
7. Discuss the significance of requirements engineering in ensuring software quality and project success. Outline and briefly describe the three fundamental activities that make up the requirements engineering process. **(2+3 marks)**

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Gat No - 5 & 6, Vill - Nanoli-Tarf Chakan, PO – Talegaon, Tah – Maval, Dist - Pune, Maharashtra – 410507

Mid-Sem Examination
(BT14503) CRYPTOGRAPHY AND NETWORK SECURITY

Course: Third Year B. Tech (Semester V) | Branch: CSE
Duration: 90 Minutes | Max. Marks: 30
Date: 29/10/2025 | Student MIS no.: 112315156

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate maximum marks.
3. Write your MIS Number on Question Paper.
4. Writing anything on question paper (except MIS No.) is not allowed.
5. Mobile phones, smart watches and programmable calculators are strictly prohibited.
6. **Attempt all parts of the same question together in a continuous sequence.**
7. **Answers written in pencil will not be evaluated.**
8. Exchange/Sharing of stationary, calculator etc. is not allowed.

NOTE: All questions are mandatory and Answers to subparts of the same question should be written CONSECUTIVELY.

Question 1: Differentiate between active and passive attacks with suitable examples. Identify the attack which is difficult to detect and prevent. Provide proper justification to support your answer. [2.5+2.5 Marks]

Question 2: Illustrate the characteristics of cryptographic systems and requirements of secure symmetric encryption systems. [5 Marks]

Question 3: Examine the significance of isolation, modularity, encapsulation, and layering concepts in designing security principles. [5 Marks]

Question 4: Consider plaintext as "Meet me at the usual place" and key $\begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$ using Hill Cipher. Demonstrate with proper steps (a) Encryption (b) Decryption [2.5 + 2.5 Marks]

Question 5: In context to Playfair Cipher, answer the following parts:

(a) Possible keys in Playfair Cipher. Support your answer with justification. [2 Marks]
(b) Assume that an attacker has knowledge of ciphertext and encryption algorithm which is Playfair Cipher but not the key. Discuss an attack strategy by which the attacker can

break the security of ciphertext. Also explain the weaknesses that permit this attack strategy to be successful. [3 Marks]

Question 6: Assume $f(x) = (x^3 + x + 1)$ and $m(x) = (x^4 + x + 1)$. Calculate $g(x)$ such that $f(x)g(x) \equiv 1 \ (mod\ m(x))$. Support your answer with step-wise calculations. Elaborate on characteristics of irreducible polynomials. [3 +2 Marks]

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Mid-term Examination
Human-Computer Interaction (BT14502)

Course: TY BTech/ MTech
Duration: 90 Minutes
Date: 28/ 10 /2025

Branch: CSE
Max. Marks: 30

Student MIS no.: 112315156

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate maximum marks.
3. Write your MIS Number on Question Paper.
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6. Exchange/Sharing of stationary, calculator etc. not allowed.

Que.1 Define HCI. Importance of HCI. Discuss the role of Psychology and AI in Interaction Design. 6 M

Que.2 Consider the scenario: You are designing a control dashboard for an air traffic control system where the operator needs to frequently switch between several buttons placed across the screen. Each button represents a different action such as "Zoom Radar", "Emergency Override", "Weather Overlay", and "Communication Channel". The layout of these buttons can significantly affect the operator's performance. 8 M
You are tasked to evaluate three different interface layouts using Fitts's Law, and determine which one is optimal for minimizing the average movement time.

Layout	Target Size (W)	Average Distance to Target (D)	Number of Targets	Weight (Usage Frequency %)
A	30	250	6	25%
B	50	400	6	40%
C	80	600	6	35%

Use the Fitts's Law, where a=0.1 sec and b=0.12 sec/bit
Write the following answers,

1. Compute the Movement Time for each layout.
2. Compute the Weighted Average Movement Time considering usage frequency.
3. Determine which layout gives the best performance (lowest effective movement time).
4. Discuss how button size vs. distance trade-offs affect.
5. Suggest one design modification that could further reduce movement time based on your calculations.

Que.3 What are the familiar senses in HCI. Discuss role of each in interaction design with example of any payment app (like BHIM, PhonePay,Gpay etc.) for different users. 6 M

Que.4 Given Scenario: You have been assigned to redesign a national government e-payment and subsidy portal used by millions of citizens across India. Current usability problems include: 10 M

1. Users abandon transactions mid-way due to confusing navigation.
2. Important notifications (like subsidy approvals) are often missed.
3. Users make repeated errors while filling forms (e.g., incorrect account numbers).
4. Some users are elderly or have low digital literacy, increasing error rates.
5. Mobile and desktop versions of the portal have inconsistent layouts, leading to confusion.

Write answers below,

A. Map the seven stages of action (Norman's Model of Interaction) to the current portal workflow.

B. Identify where the Gulf of Execution and Gulf of Evaluation is largest, citing concrete examples (e.g., form entry errors, missed notifications).

C. Identify at least 3 Gestalt principles (e.g., Proximity, Similarity, Figure-Ground, Continuity, Closure) that are violated in the current design.

********** END **********

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End Term Examination
(BT13117) Advanced computer network

Course: III Year B. Tech (Semester V)
Duration: 1.5 hours
Date: 28/10/2025

Branch: CSE 3
Max. Marks: 50

Student MIS no.: 112315156

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate maximum marks.
3. Write your MIS Number on Question Paper.
4. Writing anything on question paper (except MIS No.) is not allowed.
5. Mobile phones, smart watches and programmable calculators are strictly prohibited.
6. Exchange/Sharing of stationary, calculator etc. not allowed.

Q. 1 Describe the components of a router. Explain the functions of the control plane, data plane, and management plane. [4]

Q. 2 Explain different types of packets handled by routers. How are forwarding tables constructed and maintained? [4]

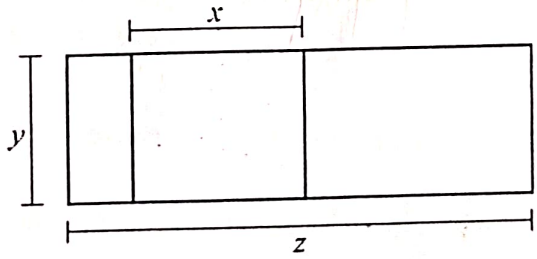
Q. 3 Discuss link characteristics with respect to bandwidth, propagation delay, and transmission time. [2]

Q. 4 Write short notes on: [4]

(a) Pipe Diagram representation of links
(b) Resource sharing in computer networks

Q. 5 Explain the Internet Design Principles. Discuss how robustness and end-to-end arguments shaped the Internet's architecture. [4]

Q. 6 Consider the pipe diagram below, with a single packet in the pipe:



[1]

Q6.1 What is the size of this packet?

i. xy ii. xz iii. yz iv. x+y v. x+z vi. y+z

vii. xy+z viii. Not enough Information

Q6.2 (1 point) How long does it take to send a packet of the same size as the packet shown? (Count from the time the first byte is sent, to the time the last byte is received.) [1]

i. xy ii. xz iii. yz iv. $x+y$ v. $x+z$ vi. $y+z$ vii. $xy+z$

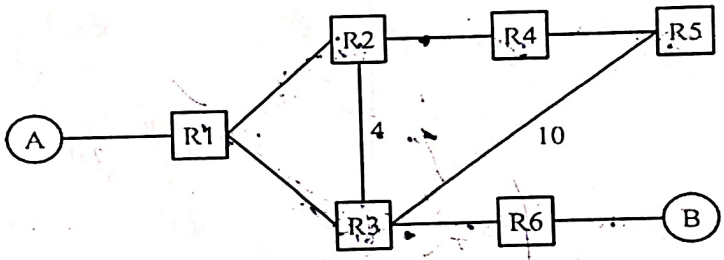
viii. Not enough information

Q6.3 How long does it take to send 7 packets, all of the same size as the packet shown? (Count from the time the first byte of the first packet is sent, to the time the last byte of the last packet is received.) [2]

i. $7xy$ ii. $7xz$ iii. $7yz$ iv. $7x+y$ v. $7x+z$ vi. $7y+z$

vii. $x+7y$ viii. Not enough information

Q. 7 Consider running the distance-vector protocol on the topology below. All unlabeled links cost 1.



At time $t = 0$, all forwarding tables are empty. At time $t = 1$, static routes are installed. At each subsequent time step, every router advertises all of its routes to all of its neighbors.

consider this scenario:

- Split horizon is enabled.
- No poison is ever sent.
- A long time later, the network has converged. After convergence, host A leaves the network.
- At time $t = 100$, R2 still has its entry for destination A (the same entry that it had at convergence).
- At time $t = 100$, all other routers have had their entries for destination A expired and deleted

Q7.1 At time $t = 103$, what table entry does each router have for destination A?

Note: $t = 103$ means you should consider three rounds of sending advertisements, receiving adver-

tisements, and updating tables. The first round ($t = 101$) starts with R2 advertising its entry for A. Some entries are filled in for you. For example, the bottom-right row says that R6's forwarding table has the entry "I can reach A with cost 7, via R3."

In the Cost column, write an integer, or ∞, or "N/A" (if the table has no entry for destination A).

In the Next-Hop column, write a router (e.g. "R1"), or "N/A" (if the entry has cost infinity, or if the table has no entry for destination A). [6]

Router	Cost to A	Next-Hop to A
R1		
R2		
R3		

Router	Cost to A	Next-Hop to A
R4	3	R2
R5		
R6	7	R3

Q7.2 Many time steps later, what happens to the routers' table entries for destination A? (Reminder: All costs 16 or greater are infinite, and all costs 15 or less are finite.) [2]

i. Every router's entry has infinite cost.

ii. Only R2 has an entry with infinite cost, and all other routers have an entry with finite cost.

iii. R1, R2, R3 have an entry with infinite cost, and R4, R5, R6 have an entry with finite cost.

iv. All routers have an entry with finite cost

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Mid-Semester Examination
(BT13507) MACHINE LEARNING

Course: T. Y. B. Tech (Semester V) + F. Y. M. Tech (Semester I) Branch: CSE and ECE
Duration: 1.5 hours Max. Marks: 30
Date: 27.10.2025 Student MIS no.: 112315156

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate maximum marks.
3. Write your MIS Number on Question Paper.
4. Writing anything on question paper (except MIS No.) is not allowed.
5. Mobile phones, smart watches, and programmable calculators are strictly prohibited.
6. Exchange/Sharing of stationary, calculator etc. not allowed.
7. **Answer all subparts of each question together with proper numbering.**

1. Answer the following: [5+2.5+2.5 = 10]

   a. Consider the following dataset. Perform K-Means clustering with K = 3, using Euclidean distance for 2 iterations. Use initial centroids at A, D, and G. Show all necessary steps.

Data Point	X	Y
A	2	10
B	2	5
C	8	4
D	5	8
E	7	5
F	6	4
G	1	2
H	4	9

   b. What are the criteria for convergence in K-Means clustering? In the above example, has the clustering converged? – Discuss briefly.

   c. Explain the advantages and disadvantages of K-Means clustering.

2. Answer the following: [5+5+2+3=15]

a. Derive the closed form solution for simple linear regression using Cramer's Rule for matrices. Show all steps of derivation with reasons clearly.

b. Find the equation of the best fit line for the following data points using the derived formula. Show all steps clearly.

Hours of Study (X)	1	2	3	4	5	6
GPA (Y)	1.2	1.8	2.6	3.2	3.8	4

c. Use this equation to predict the GPAs for Hours of Study=7 and Hours of Study = 13. Show all steps clearly.

d. What are the different types of Gradient Descent? Explain each type in brief. What are their advantages and disadvantages?

3. Answer the following: [2+3 = 5]

a. What is One-Hot encoding? Explain with an example.

b. Discuss the differences between Normalization and Standardization. For the following data, find the normalized and standardized values for the feature "Salary". Show all steps clearly.

Person	Age	Salary (INR)
1	44	73000
2	27	17000
3	30	80000
4	38	75000
5	40	1,1500

---------- ALL THE BEST ----------

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Indian Institute of Information Technology, Pune
Mid Term Examination 2025-2026
(BT14401) Linear and Nonlinear Optimization

Course : F .Y. M. Tech. (Semester I, CSE) and T. Y. B. Tech. (Semester V, CSE & ECE)
Duration : 90 minutes — Max. Marks : 30
Date : 27/10/2025 — Student MIS No.: 112315156

Instructions:
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7. Assume suitable data, if necessary.

Q 1. Answer the following questions.

(a) State the fundamental theorem of duality. [01]

(b) When do we get degeneracy in a Linear Programming Problem (LPP) while using the revised simplex method? How is it resolved? [02]

(c) Examine whether the **epigraph** of a convex function $f(x)$, [02]

$$\text{epi}(f) = \{(x,t) \mid f(x) \leq t\},$$

is convex or not. Justify your answer. Also, find its convex hull.

(d) Consider the following LPP and answer the questions below.

$$Min\ (Z) = 3x_1 + 2x_2 - 2x_3$$

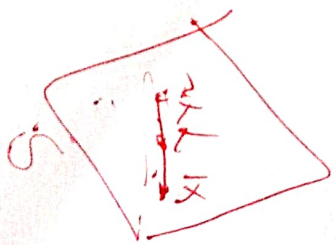
subject to: $3x_1 + x_2 - 2x_3 \geq 4$, $3x_1 + 4x_2 = 6$; $x_2, x_3 \geq 0, x_1$ is unresticed in sign.

i. Solve using the Big M method. [07]

ii. Write down the dual of the above LPP. (Show all steps in details) [03]

Q 2. Answer the following questions.

(a) A software firm has 3 programmers and 2 testers available per week. [03]
Project P_1 requires 1 programmer and 1 tester per week.
Project P_2 requires 2 programmers and 1 tester per week.
Profit per week: P_1 = \$500, P_2 = \$800.
Formulate a Linear Programming Problem (LPP) to maximize weekly profit while meeting resource constraints. Solve using the graphical method.



(b) i. Discuss the effect on the **optimal basic feasible solution (BFS)** of the Linear Programming problem: [06]

$$\text{Maximize}(Z) = c^T x, \quad \text{subject to } Ax = b,\ x \geq 0,$$

when a **cost coefficient** in the objective function is changed.

ii. Consider the following LP problem with optimal simplex table: [02]

$Max(Z) = 2x_1 + 3x_2 + 4x_3$

Subject to; $x_1 + 2x_2 + 3x_3 \leq 11$; $2x_1 + 3x_2 + 2x_3 \leq 10$; $x_1, x_2, x_3 \geq 0$.

B	C_B	X_B	x_1	x_2	x_3	s_1	s_2
x_3	4	3	0	1/4	1	1/2	-1/4
x_1	2	2	1	5/4	0	-1/2	3/4
$Max(Z) =$	16	Δ_j	0	1/2	0	1	1/2

Discuss the effect on the optimal basic feasible solution by adding a new variable x_4 with column coefficients $(2, 3)^T$ and coefficient 2 in the objective function.

(c) Determine the inverse of the basic matrix at the second iteration while solving the following Linear Programming Problem (LPP) using the Revised Simplex Method. Show all steps in detail. [04]

$Max(Z) = 5x_1 + 10x_2 + 8x_3$

Subject to; $3x_1 + 5x_2 + 2x_3 \leq 60$;

$4x_1 + 4x_2 + 4x_3 \leq 72$

$x_1, x_2, x_3 \geq 0$

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